ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/332511537

Validity of a Muscle Specific Method to Evaluate the Anaerobic Threshold in Exercised Muscles

Article *in* Human Physiology · February 2019

DOI: 10.1134/S0362119719020087

CITATIONS
3

READS
287

6 authors, including:

Daniil Popov
Institute of Biomedical Problems of the Russian Academy of Sciences
115 PUBLICATIONS **2,254** CITATIONS

SEE PROFILE

Anatoly S. Borovik
Institute of Biomedical Problems, Russian Academy of Sciences
80 PUBLICATIONS **475** CITATIONS

SEE PROFILE

All content following this page was uploaded by Daniil Popov on 18 April 2019.

The user has requested enhancement of the downloaded file.

ISSN 0362-1197, Human Physiology, 2019, Vol. 45, No. 2, pp. 174–179. © Pleiades Publishing, Inc., 2019.
Russian Text © D.V. Popov, S.Yu. Kuznetsov, E.A. Orlova, A.P. Sharova, A.S. Borovik, O.L. Vinogradova, 2019, published in Fiziologiya Cheloveka, 2019, Vol. 45, No. 2, pp. 70–77.

Validity of a Muscle Specific Method to Evaluate the Anaerobic Threshold in Exercised Muscles[1]

D. V. Popov[a, b, *], S. Yu. Kuznetsov[a], E. A. Orlova[a], A. P. Sharova[a], A. S. Borovik[a], and O. L. Vinogradova[a, b]

[a] *Institute of Biomedical Problems of the Russian Academy of Sciences, Moscow, Russia*
[b] *Moscow State University, Moscow, Russia*
**e-mail: danil-popov@yandex.ru*

Received August 7, 2018; revised September 23, 2018; accepted November 1, 2018

Abstract— The goal of this study was to describe and validate a muscle specific method to evaluate the anaerobic threshold in a working muscle based on the simultaneous measurement of EMG activity and the deoxyhemoglobin content ($AT_{HHb\text{-}EMG}$). The study involved males with different fitness levels. During the cycling (n = 40) and ski double poling (n = 9) incremental ramp tests, blood lactate concentration, and muscle deoxyhemoglobin content and EMG activity were measured. Some participants were involved in the cycling test-retest study (n = 11). In cycling and double poling tests, close and significant correlations (r = 0.89 – 0.92, P < 0.002) were found between lactate threshold (a marker of the anaerobic threshold at the organism level) and the $AT_{HHb\text{-}EMG}$ (a marker of the AT at the working muscle level). The coefficient of variation of the $AT_{HHb\text{-}EMG}$ in the cycling test-retest was low (~3%). The muscle specific $AT_{HHb\text{-}EMG}$ demonstrates low variability and is appropriate to detect the fitness level and training-induced increase in aerobic performance in a working muscle. The emergence on the market of miniature EMG amplifiers and near-infrared spectrometers opens wide possibilities for using our method in laboratory studies and in field tests.

Keywords: exercise, muscle, anaerobic threshold, EMG, NIRS

DOI: 10.1134/S0362119719020087

Evaluation of aerobic performance is very important in the fields of sport, fitness, and medicine. Power at the anaerobic threshold (AT), as evaluated in an incremental exercise test, is widely used to predict aerobic performance and to select the optimal load in exercise training. Many studies show that the AT is an informative and robust predictor of whole-body aerobic performance [1–5]. Several methods to detect the AT during an incremental test have been described in the literature. Most of them use systemic parameters (blood lactate concentration, pulmonary ventilation, and O_2 consumption and CO_2 production rates, etc.) to detect the AT [1, 6–8]. Therefore, these methods can only be correctly applied to the work of a large muscle mass (running, cycling, skiing, etc.).

During low- and moderate-intensity aerobic exercise, low-threshold motoneurons innervating type I muscle fibers (MFs) are mainly recruited in human muscles. A gradual increase in exercise intensity is associated with the recruitment of high-threshold motoneurons and type II MFs [9], which have a lower oxidative capacity than type I MFs. The increase in intensity leads to the progressive accumulation of various metabolites in working muscles, especially in type II MFs, development of fatigue, and consequently an increase in the recruitment rate of MFs. These processes provide an opportunity to detect the AT during an incremental exercise test in a working muscle. Indeed, "inflection points" in the phosphocreatine content and pH in a working muscle [10, 11], in integrated electromyography (EMG) activity (a marker of motor unit recruitment) [12–14] and in the muscle deoxyhemoglobin (HHb) content [15] (an index related to the muscle O_2 extraction) were found during an incremental test using nuclear magnetic resonance spectroscopy, EMG, and near-infrared spectroscopy, respectively. These approaches allow evaluation of the muscle specific AT in a working muscle during incremental exercise of either a large or small muscle mass. However, there are notable limitations to the methods. Measurement of the phosphocreatine content and pH in a working muscle using nuclear magnetic resonance spectroscopy is expensive and inconvenient. Inflection point in EMG activity (EMG threshold) is not necessarily present in an incremental test [16, 17] and demonstrates relatively low the test-retest reliability [16–20] compare to conventional methods [21]. Several approaches were proposed to evaluate the muscle specific anaerobic threshold by near-infrared spec-

[1] The article was translated by the authors.

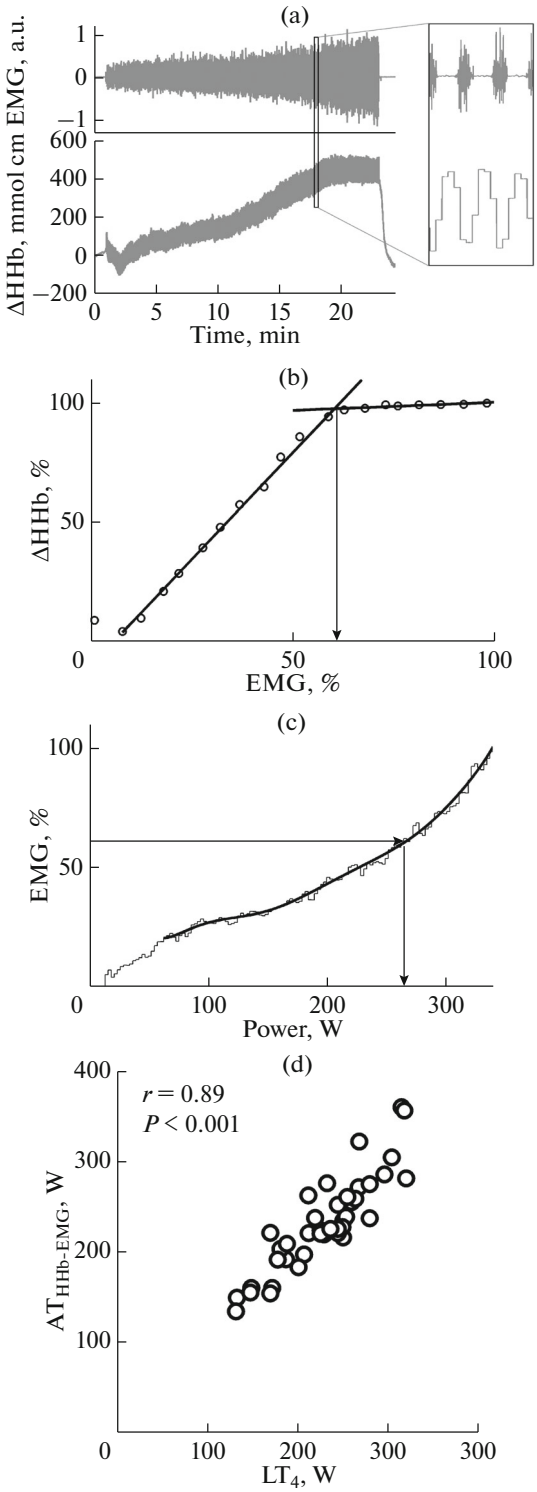

Fig. 1. The muscle specific anaerobic threshold ($AT_{HHb\text{-}EMG}$) in the m. vastus lateralis during the cycling incremental ramp test (load increment rate, 15 W/min) was evaluated based on EMG activity and changes in the deoxyhemoglobin (HHb) content. Representative data are shown of native EMG and HHb signals (a). The inflection point in the relationship between normalized HHb and EMG was evaluated using the V-slope method (b), and then power at the $AT_{HHb\text{-}EMG}$ was evaluated (c). The $AT_{HHb\text{-}EMG}$ significantly correlates with LT_4 in the cycling test (d); $n = 40$.

troscopy [22–24], in particular using the muscle HHb content [15, 25–29], but the methods have not been well validated.

Recently, we described a muscle specific method to detect the AT in a working muscle simultaneously using both HHb and EMG ($AT_{HHb\text{-}EMG}$) [30]. In the current study, the algorithm for detection of the $AT_{HHb\text{-}EMG}$ was modified (see Methods) to increase its robustness and then the method was validated. For this purpose, we (1) compared the $AT_{HHb\text{-}EMG}$ with the well-described and widely used lactate threshold in cycling and double poling incremental tests; (2) evaluated the test-retest reliability of the $AT_{HHb\text{-}EMG}$.

METHODS

Algorithm for estimation of muscle specific $AT_{HHb\text{-}EMG}$ during incremental ramp test. EMG activity and the HHb content in a working muscle are to be recorded continuously during an incremental ramp test (see below; Fig. 1a). EMG signals were rectified, and then mean values of *EMG* and *HHb* averaged over 20 s were calculated (*mEMG* and *mHHb*). The mean values were then normalized using the following formulas:

$$nEMG = (mEMG - \max[mEMG]) / (\max[mEMG] - \min[mEMG]), \quad (1)$$

$$nHHb = (mHHb - \max[mHHb]) / (\max[mHHb] - \min[mHHb]), \quad (2)$$

where max(*mEMG*) and max(*mHHb*) are maximal values of *mEMG* and *mHHb* in the incremental test, respectively. Sets of $nEMG_k$ values, lying in intervals $[k \ldots k + 1] \times 0.05$, where k was changed from 0 to 19, and sets of corresponding $nHHb_k$ were collected, mean values were calculated, and then the function $nHHb_k(nEMG_k)$ was constructed (Figs. 1b and 2a). Couples ($nEMG_k$, $nHHb_k$) were divided into two groups with k lying in the interval $[1 \ldots i]$ and $[i+1 \ldots 20]$, where i increased from 3 to 18, and an inflection point was found using the V-slope method [31] (Figs. 1b and 2a). The abscissa of the intersection of the lines with the minimal sum of squared residuals determined the *nEMG* value in the $AT_{HHb\text{-}EMG}$. The power (or time) corresponding to the $AT_{HHb\text{-}EMG}$ was determined on the plot of *nEMG* vs. *power* (or *time*), approximated with a polynomial of order 5 (Figs. 1c, 2b, 2c). Data processing and calculation of the $AT_{HHb\text{-}EMG}$ were performed with MATLAB (MathWorks, USA) programs.

Comparison of the muscle specific $AT_{HHb\text{-}EMG}$ with the lactate threshold. The lactate threshold evaluated by a fixed lactate level of 4 mmol/L in blood (LT_4) is a very informative and robust predictor of aerobic performance [5]. Therefore, we compared the well-described LT_4 with the muscle specific $AT_{HHb\text{-}EMG}$ in

different incremental exercises with a large muscle mass (cycling and double poling).

Cycling incremental ramp test. The study involved 40 males (median age, 26 years (interquartile range, 20–35 years); weight, 70 kg (65–85 kg)) with different fitness levels, from untrained men to amateur endurance athletes (cross county skiers, runners, and cyclists). During the first visit to the laboratory, all participants were familiarized with the test procedures.

Two to three days later, each participant performed an incremental ramp test on an electromagnetic bicycle ergometer (Ergoselect 200, Ergoline, Germany). The initial load, load increment rate, and revolution rate were 0 W, 15 W/min, and 60–70 rpm, respectively. Each participant exercised until exhaustion. During the test, a sample of capillary blood (20 μL) was taken from a fingertip of the subject every 2 min and immediately after exhaustion. Lactate concentrations were determined using a Biosen C-line analyzer (EKF Diagnostics, Germany). EMG activity was continuously measured in the middle part of the m. vastus lateralis using a CP511 amplifier (Grass Telefactor, USA) and standard Ag/AgCl electrodes. The signal was amplified 1000-fold and passed through a 3–300 Hz bandpass filter. At the site of electrode attachment, the skin was shaved, cleaned with an abrasive pad, and then treated with alcohol. Changes in the concentration of HHb (ΔHHb) were measured at a frequency of 6 Hz using a NIRO-200 spectrophotometer (Hamamatsu Photonics K.K., Japan). The infrared source and receiver were placed 40 mm from each other, were located close to the EMG electrodes, and were fixed on the skin with tape and an elastic bandage. The analog signals from both devices were digitized at a frequency of 1000 Hz using an E440 analog-to-digital converter (L-Card, Russia) and recorded using PowerGraph software (DISoft, Russia).

Double poling incremental step test. The study involved nine untrained males (age, 26 years (21–35 years); weight, 73 kg (66–74 kg)). All subjects were familiarized with the test procedures and then completed an incremental ramp test until exhaustion on a double poling ski ergometer SkiErg (Concept2, USA). The initial load and load increment rate were 25 W and 10 W/min, respectively. Each subject closely maintained the required target load through visual feedback.

Two to three days later, each participant performed two incremental tests until exhaustion in a randomized order, namely, an incremental ramp test and an incremental step test. The rest period between the tests was 1 day. The ramp test was performed as described above. In the step test, the initial load, step duration, and load increase were 25 W, 2 min, and 20 W, respectively. Importantly, the mean load increment rate in both tests was 10 W/min.

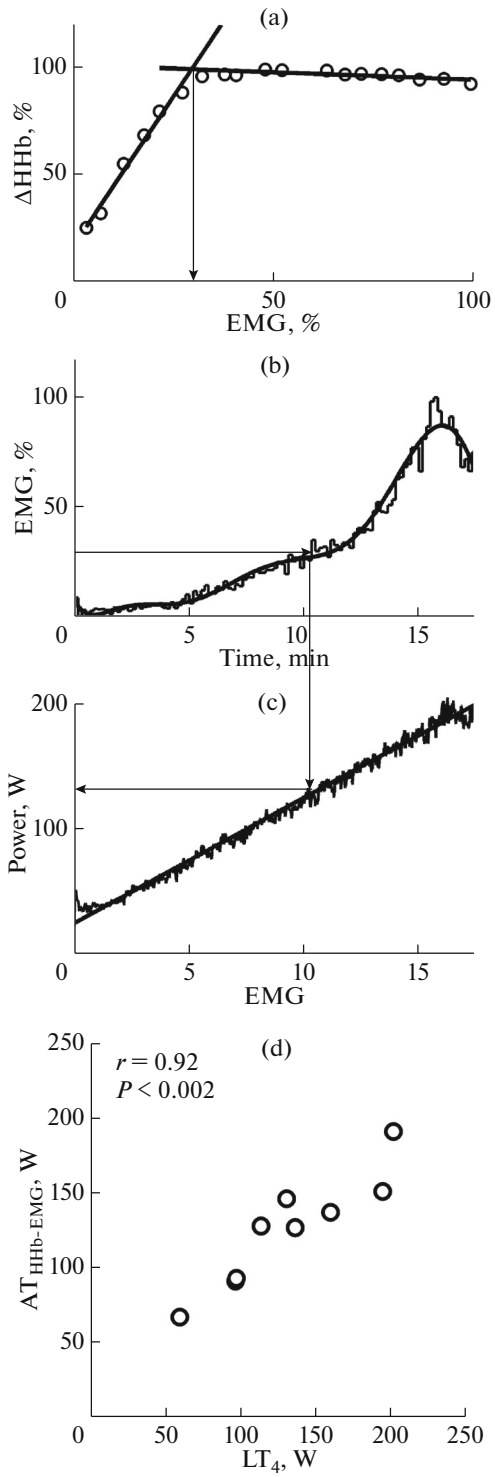

Fig. 2. The muscle specific anaerobic threshold ($AT_{HHb\text{-}EMG}$) in the m. latissimus dorsi during the double poling incremental ramp test was evaluated based on EMG activity and changes in the deoxyhemoglobin (HHb) content. The inflection point in the relationship between normalized HHb and EMG was evaluated using the V-slope method (a), and then time (b) and power (c) at the $AT_{HHb\text{-}EMG}$ were evaluated. The $AT_{HHb\text{-}EMG}$ in the double poling ramp test significantly correlates with LT_4 in the double poling step test (d), $n = 9$. The mean load increment rate was 10 W/min in both tests.

Table 1. Physiological data and coefficients of variation (CV) in the incremental cycling test and retest

Physiological data	W_{max}, W	La_{peak}, mmol/L	LT_4, W	$AT_{HHb\text{-}EMG}$, W
Test	257 (246–311)	9.0 (8.0–10.3)	201 (153–244)	219 (166–236)
Retest	262 (251–314)	9.1 (8.8–10.2)	188 (170–249)	221 (158–234)
P-level	0.08	0.52	0.82	0.52
CV, %	1.2 (0.7–1.9)	7.6 (4.1–12.2)	2.4 (1.0–8.0)	3.1 (0.8–5.4)

W_{max}—maximal aerobic power, La_{peak}—capillary blood lactate concentration at exhaustion, LT_4—lactate threshold, $AT_{HHb\text{-}EMG}$—anaerobic threshold in the m. vastus lateralis. Each value represents the median and interquartile range, $n = 11$.

During the ramp test, EMG activity and ΔHHb in the distal half of the m. latissimus dorsi were measured as described above for calculation of the $AT_{HHb\text{-}EMG}$. During the step test, capillary blood samples were taken from a fingertip after each step during a short rest period (10–15 s) to calculate LT_4.

Test-retest reliability of the $AT_{HHb\text{-}EMG}$. Eleven males (age, 27 years (24–34 years); weight, 72 kg (70–79 kg)) who performed the cycling incremental ramp test repeated the test within 1 wk. The same parameters were measured in the retest.

Statistics. Some data had a non-normal distribution, and thus were expressed as the median and interquartile range. The Wilcoxon matched-pairs signed-rank test was used to compare related measurements. Correlations were assessed using Spearman's correlation coefficient. The level of significance was set at 0.05. The coefficient of variation (CV) was calculated as the ratio of the standard deviation to the mean expressed as a percentage.

RESULTS

Comparison of the $AT_{HHb\text{-}EMG}$ with the lactate threshold. The power at $AT_{HHb\text{-}EMG}$ and LT_4 were significantly correlated in both the cycling ($r = 0.89$, $P < 0.001$, Fig. 1d) and double poling ($r = 0.92$, $P < 0.002$, Fig. 2d) tests.

Test-retest reliability of the $AT_{HHb\text{-}EMG}$. The power at $AT_{HHb\text{-}EMG}$ and at LT_4, maximal aerobic power (W_{max}), and blood lactate content at exhaustion (La_{peak}) did not differ between the incremental cycling test and retest (Table 1). The coefficient of variation (CV) was smallest for W_{max} and largest for La_{peak}. CVs did not differ between the $AT_{HHb\text{-}EMG}$ and LT_4, and were low (3.1% (0.8–5.4%) and 2.4% (1.0–8.0%), respectively) (Table 1).

DISCUSSION

The current study validated the method to detect the AT in a working muscle. The principle of the method is based on the following background: (1) towards the end of the incremental test, the progressive accumulation of metabolites in a working skeletal muscle leads to development of fatigue, an increased recruitment rate of motor units, and consequently a non-linear increase in normalized EMG activity; and (2) the muscle HHb content, an index related to the muscle O_2 extraction [32], increases during the incremental test, but the rate of increase slows down towards the end of the test. The relationship between the HHb content and EMG activity demonstrates a well-detected inflection point (Figs. 1b and 2a). This inflection point was well-detectable in the m. vastus lateralis during the cycling and one-legged incremental tests and in the m. latissimus dorsi during the double poling incremental test. Importantly, the m. vastus lateralis and m. latissimus dorsi play a dominant role in cycling and double poling, respectively. In both tests, close and significant correlations were found between LT_4 (a marker of the AT at the organism level) and the $AT_{HHb\text{-}EMG}$ (a marker of the AT at the working muscle level). We speculate that the correlation may be less significant in the non-dominant muscle and that the $AT_{HHb\text{-}EMG}$ in the muscle may differ from LT_4. Noteworthy, the $AT_{HHb\text{-}EMG}$ may give an opportunity to detect less or more trained muscles during the incremental work of a large muscle mass.

Finally, an important characteristic of each method is variability. In previous studies, the CV for LT_4 was 1.4% to 5.9% [21, 33–35]; the values are comparable with the CV for LT_4 in our study (CV, 2.4%). Variability of the $AT_{HHb\text{-}EMG}$ in the cycling test-retest was low (CV, 3.1%) and did not differ from that of the LT_4.

There is a notable limitation to our method. Near-infrared light penetrates the tissue at least deep enough to reach half the distance between the light source and the detector [36] (40 mm in our study). Subcutaneous adipose tissue thickness more than 10 mm may mask muscle-related changes in the HHb content [36]. Additionally, the changes in skin perfusion and the effect of scattering changes in tissue during an incremental test may affect the HHb content [37].

CONCLUSIONS

In conclusion, many methods for detecting the AT during whole-body exercise are described in the literature. Several studies to evaluate the muscle specific

AT in a working muscle utilized EMG activity or HHb, parameters related to motor unit recruitment and muscle oxygen extraction respectively. The AT evaluated by these approaches is not necessary present in an incremental test [15–17] and demonstrates relatively low the test-retest reliability [16–21]. In the current study, to increase robustness of the method to evaluate the muscle specific AT in a working muscle both EMG activity and the HHb content were used. The $AT_{HHb\text{-}EMG}$ demonstrates low variability and significantly correlates with the well-described LT_4. Muscle specific $AT_{HHb\text{-}EMG}$ may be used for detecting less or more trained muscles during work of a large muscle mass. Importantly, the appearance on the market of low-cost wireless diminutive EMG amplifiers and near-infrared spectrometers opens wide possibilities for using our method in laboratory studies and in field tests.

ACKNOWLEDGMENTS

The study was performed according to the Plan for Fundamental Research of the SRC RF Institute of Biomedical Problems RAS and was funded by RFBR according to the research project no. 17-00-00308 (17-00-00242).

COMPLIANCE WITH ETHICAL STANDARDS

Conflict of interests. The authors declare that they have no conflict of interest.

Statement of compliance with standards of research involving humans as subjects. The study was approved by the Human Ethics Committee of the Institute of Biomedical Problems and complied with the guidelines set forth in the Declaration of Helsinki (World Medical Association). All participants gave their written consent to participate in this study.

REFERENCES

1. Farrell, P.A., Wilmore, J.H., Coyle, E.F., et al., Plasma lactate accumulation and distance running performance, *Med. Sci. Sports*, 1979, vol. 11, no. 4, p. 338.
2. Komi, P.V., Ito, A., Sjodin, B., et al., Muscle metabolism, lactate breaking point, and biomechanical features of endurance running, *Int. J. Sports Med.*, 1981, vol. 2, no. 3, p. 148.
3. Loat, C.E. and Rhodes, E.C., Relationship between the lactate and ventilatory thresholds during prolonged exercise, *Sports Med.*, 1993, vol. 15, no. 2, p. 104.
4. Mader, A., Evaluation of the endurance performance of marathon runners and theoretical analysis of test results, *J. Sports Med. Phys. Fitness* 1991, vol. 31, no. 1, p. 1.
5. Faude, O., Kindermann, W., and Meyer, T., Lactate threshold concepts: how valid are they? *Sports Med.*, 2009, vol. 39, no. 6, p. 469.
6. Beaver, W.L., Wasserman, K., and Whipp, B.J., A new method for detecting anaerobic threshold by gas exchange, *J. Appl. Physiol.*, 1986, vol. 60, no. 6, p. 2020.
7. Beaver, W.L., Wasserman, K., and Whipp, B.J., Improved detection of lactate threshold during exercise using a log-log transformation, *J. Appl. Physiol.*, 1985, vol. 59, no. 6, p. 1936.
8. Wasserman, K. and McIlroy, M.B., Detecting the threshold of anaerobic metabolism in cardiac patients during exercise, *Am. J. Cardiol.*, 1964, vol. 14, p. 844.
9. Gollnick, P.D., Piehl, K., and Saltin, B., Selective glycogen depletion pattern in human muscle fibers after exercise of varying intensity and at varying pedalling rates, *J. Physiol.*, 1974, vol. 241, no. 1, p. 45.
10. Marsh, G.D., Paterson, D.H., Thompson, R.T., and Driedger, A.A., Coincident thresholds in intracellular phosphorylation potential and pH during progressive exercise, *J. Appl. Physiol.*, 1991, vol. 71, no. 3, p. 1076.
11. Zanconato, S., Buchthal, S., Barstow, T.J., and Cooper, D.M., 31P-magnetic resonance spectroscopy of leg muscle metabolism during exercise in children and adults, *J. Appl. Physiol.*, 1993, vol. 74, no. 5, p. 2214.
12. Viitasalo, J.T., Luhtanen, P., Rahkila, P., and Rusko, H., Electromyographic activity related to aerobic and anaerobic threshold in ergometer bicycling, *Acta Physiol. Scand.*, 1985, vol. 124, no. 2, p. 287.
13. Nagata, A., Muro, M., Moritani, T., and Yoshida, T., Anaerobic threshold determination by blood lactate and myoelectric signals, *Jpn. J. Physiol.*, 1981, vol. 31, no. 4, p. 585.
14. Moritani, T., Takaishi, T., and Matsumoto, T., Determination of maximal power output at neuromuscular fatigue threshold, *J. Appl. Physiol.*, 1993, vol. 74, no. 4, p. 1729.
15. Grassi, B., Quaresima, V., Marconi, C., et al., Blood lactate accumulation and muscle deoxygenation during incremental exercise, *J. Appl. Physiol.*, 1999, vol. 87, no. 1, p. 348.
16. Taylor, A.D. and Bronks, R., Reproducibility and validity of the quadriceps muscle integrated electromyogram threshold during incremental cycle ergometry, *Eur. J. Appl. Physiol. Occup. Physiol.*, 1995, vol. 70, no. 3, p. 252.
17. Hug, F., Laplaud, D., Lucia, A., and Grelot, L., EMG threshold determination in eight lower limb muscles during cycling exercise: a pilot study, *Int. J. Sports Med.*, 2006, vol. 27, no. 6, p. 456.
18. Lucia, A., Sanchez, O., Carvajal, A., and Chicharro, J.L., Analysis of the aerobic-anaerobic transition in elite cyclists during incremental exercise with the use of electromyography, *Br. J. Sports Med.*, 1999, vol. 33, no. 3, p. 178.
19. Pavlat, D.J., Housh, T.J., Johnson, G.O., et al., An examination of the electromyographic fatigue threshold test, *Eur. J. Appl. Physiol Occup. Physiol.*, 1993, vol. 67, no. 4, p. 305.
20. Mahmutovic, S., Sprout, E.Y., Fontaine, J.C., et al., Test-retest reliability of the electromyographic fatigue threshold for cycle ergometry, *Muscle Nerve*, 2016, vol. 53, no. 5, p. 803.

21. Hopkins, W.G., Schabort, E.J., and Hawley, J.A., Reliability of power in physical performance tests, *Sports Med.*, 2001, vol. 31, no. 3, p. 211.
22. Bhambhani, Y.N., Buckley, S.M., and Susaki, T., Detection of ventilatory threshold using near infrared spectroscopy in men and women, *Med. Sci. Sports Exercise*, 1997, vol. 29, no. 3, p. 402.
23. Belardinelli, R., Barstow, T.J., Porszasz, J., and Wasserman, K., Changes in skeletal muscle oxygenation during incremental exercise measured with near infrared spectroscopy, *Eur. J. Appl. Physiol. Occup. Physiol.*, 1995, vol. 70, no. 6, p. 487.
24. Soller, B.R., Yang, Y., Lee, S.M., et al., Noninvasive determination of exercise-induced hydrodgen ion threshold through direct optical measurement, *J. Appl. Physiol.*, 2008, vol. 104, no. 3, p. 837.
25. Bellotti, C., Calabria, E., Capelli, C., and Pogliaghi, S., Determination of maximal lactate steady state in healthy adults: can NIRS help? *Med. Sci. Sports Exercise*, 2013, vol. 45, no. 6, p. 1208.
26. Osawa, T., Kime, R., Hamaoka, T., et al., Attenuation of muscle deoxygenation precedes EMG threshold in normoxia and hypoxia, *Med. Sci. Sports Exercise*, 2011, vol. 43, no. 8, p. 1406.
27. Fontana, F.Y., Keir, D.A., Bellotti, C., et al., Determination of respiratory point compensation in healthy adults: Can non-invasive near-infrared spectroscopy help? *J. Sci. Med. Sport*, 2015, vol. 18, no. 5, p. 590.
28. Keir, D.A., Fontana, F.Y., Robertson, T.C., et al., Exercise intensity thresholds: identifying the boundaries of sustainable performance, *Med. Sci. Sports Exercise*, 2015, vol. 47, no. 9, p. 1932.
29. Wang, L., Yoshikawa, T., Hara, T., et al., Which common NIRS variable reflects muscle estimated lactate threshold most closely? *Appl. Physiol. Nutr. Metab.*, 2006, vol. 31, no. 5, p. 612.
30. Kuznetsov, S.Yu., Popov, D.V., Borovik, A.S., and Vinogradova, O.L., Determination of aerobic–anaerobic transition in the working muscle using EMG and near-infrared spectroscopy data, *Hum. Physiol.*, 2015, vol. 41, no. 5, p. 548.
31. Jones, R.H. and Molitoris, B.A., A statistical method for determining the breakpoint of two lines, *Anal. Biochem.*, 1984, vol. 141, no. 1, p. 287.
32. Grassi, B. and Quaresima, V., Near-infrared spectroscopy and skeletal muscle oxidative function in vivo in health and disease: a review from an exercise physiology perspective, *J. Biomed. Opt.*, 2016, vol. 21, no. 9, p. 091313.
33. Jensen, K. and Johansen, L., Reproducibility and validity of physiological parameters measured in cyclists riding on racing bikes placed on a stationary magnetic brake, *Scand. J. Med. Sci. Sports*, 1998, vol. 8, no. 1, p. 1.
34. Pfitzinger, P. and Freedson, P.S., The reliability of lactate measurements during exercise, *Int. J. Sports Med.*, 1998, vol. 19, no. 5, p. 349.
35. Hoefelmann, C.P., Diefenthaeler, F., Costa, V.P., et al., Test-retest reliability of second lactate turnpoint using two different criteria in competitive cyclists, *Eur. J. Sport Sci.*, 2015, vol. 15, no. 4, p. 265.
36. Homma, S., Fukunaga, T., and Kagaya, A., Influence of adipose tissue thickness on near infrared spectroscopic signals in the measurement of human muscle, *J. Biomed. Opt.*, 1996, vol. 1, p. 418.
37. Ferreira, L.F., Hueber, D.M., and Barstow, T.J., Effects of assuming constant optical scattering on measurements of muscle oxygenation by near-infrared spectroscopy during exercise, *J. Appl. Physiol.*, 2007, vol. 102, no. 1, p. 358.

View publication stats